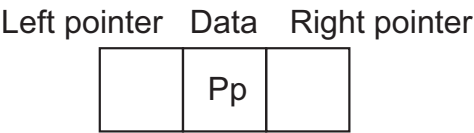


11 (a) A linked list of nodes is used to store an ordered list of strings. Each node consists of the data, a left pointer and a right pointer.

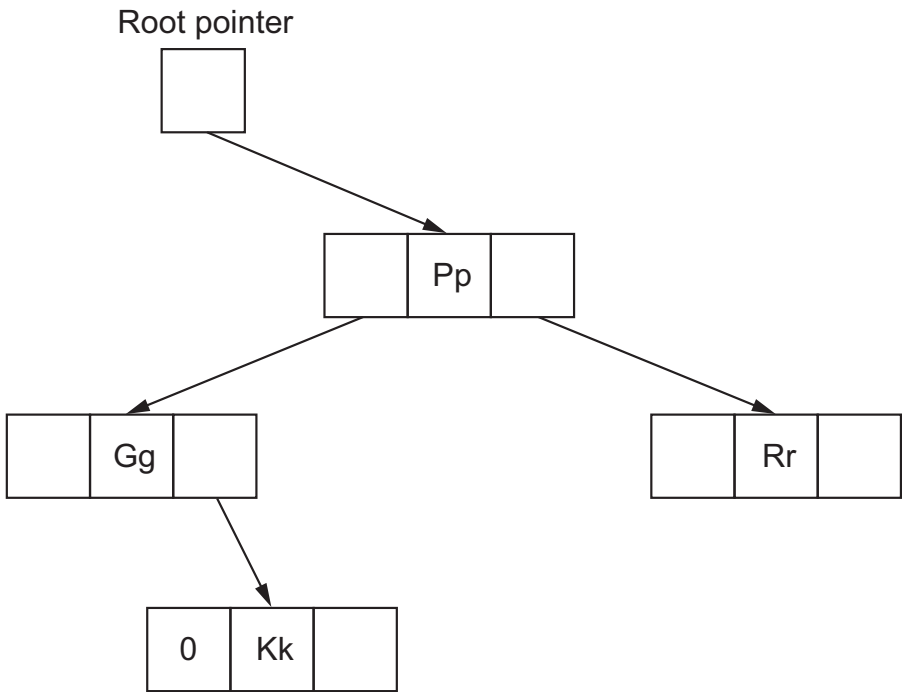


The linked list will be organised as a binary tree.

0 is used to represent a null pointer.

Complete the binary tree, including null pointers, to show how the data will be organised after the following strings have been added:

Aa, Mm, Ss, Xx



[4]

(b) A binary tree can be used to implement recursion.

Identify **one** feature of an algorithm that makes it beneficial to use recursion.
Give **one** example of an application that could use a recursive algorithm.

Feature

.....

Example

[2]